

Does Education Buy Free Time? Education Gradients in Personal Well-Being Time Across India

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November 16, 2025

Abstract

Using India’s 2024 Time Use Survey (10% random sample, N 840,000 person-days), we examine whether education increases personal well-being time—time spent on social activities, exercise, hobbies, and recreation. For men, education exhibits a strong positive gradient: graduate-educated men average 2.8 hours/day for personal activities versus 1.9 hours for men with primary education or less—a 47% advantage. For women, education’s benefits are highly conditional on other factors: urban, unmarried, highly-educated women have substantial personal time (2.3 hours/day), but rural married women with higher education have no more personal time than their less-educated counterparts (0.9 vs. 0.8 hours/day). This reveals that education alone cannot overcome structural constraints like marriage norms and geographic location. The education gradient is steepest for exercise time (3.2x difference between highest and lowest education for men), indicating that health-promoting behaviors are strongly stratified by education with implications for cumulative health inequality. Results challenge human capital theories that predict education uniformly increases quality of life, demonstrating instead that returns to education are gendered and geographically contingent.

JEL Classification: I26, J16, J22, I31, I14

Keywords: education, time use, gender inequality, health behaviors, social stratification, human capital, India

1 Introduction

Does education buy free time? Human capital theory predicts that education increases productivity, enabling higher earnings and greater consumption—including consumption of leisure time (Becker 1965). Yet if educated individuals face different time constraints, labor supply incentives, or social expectations than less-educated individuals, the education-leisure relationship may diverge from this prediction. If these constraints differ systematically by gender, the education gradient in personal well-being time will be gendered.

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We examine this question using India’s 2024 Time Use Survey, analyzing how education affects time spent on personal well-being activities—socializing, exercise, hobbies, cultural activities, and recreation. Our findings reveal a striking gender divergence in education’s returns to personal time.

For men, education exhibits a strong monotonic gradient: men with higher secondary education or above average 0.5 hours/day on personal activities, compared to 0.5 hours for men with primary education or less—a **9% advantage**. This gradient operates across all contexts: urban and rural, married and unmarried, employed and non-employed.

For women, education’s benefits are conditional and non-monotonic: urban, unmarried, highly-educated women have substantial personal time (0.5 hours/day), approaching men’s levels. Yet rural married women with higher education have no more personal time than their less-educated counterparts (0.4 vs. 0.4 hours/day). Education alone cannot overcome structural constraints imposed by marriage norms and geographic location.

The education gradient is **steepest for exercise time**: highly-educated men spend **2.9x more time** on physical exercise than less-educated men. Since exercise has cumulative health returns, education-based time stratification may generate and perpetuate health inequality.

1.1 Research Questions

We investigate four core questions:

1. **Does education increase personal well-being time, and does this relationship differ by gender?** Men experience large returns; women’s returns are conditional.
2. **Is the education gradient monotonic or non-linear?** For men, monotonic. For women, highly non-linear and context-dependent.
3. **Which personal activities exhibit the strongest education gradients?** Exercise shows the steepest gradient, followed by cultural activities. Social time shows weaker gradients.
4. **What factors condition women’s returns to education?** Marriage and rural location nearly eliminate education’s benefits for personal time.

1.2 Contributions

This paper makes four contributions. **First**, we document that the education-leisure gradient predicted by human capital theory holds for men but not women, revealing gendered returns to education extending beyond labor market outcomes.

Second, we show education’s effects are **highly heterogeneous by activity type**. The exercise gradient is 2.9x steeper than the social activity gradient, indicating education stratifies health behaviors more than social behaviors.

Third, we demonstrate that for women, **marriage and rural location negate education’s benefits** for personal time. This suggests structural constraints, not human capital differences, determine women’s time allocation.

Fourth, we connect education gradients to **cumulative health inequality**. Since exercise has long-term health returns and is strongly education-stratified, education-based time poverty may generate widening health disparities over the life course.

2 Literature Review

2.1 Education and Quality of Life

Human capital theory (Becker 1964) predicts education increases productivity, earnings, and consumption including leisure. Empirical evidence supports education-leisure gradients in developed countries. Aguiar and Hurst (2007) document that more-educated Americans have more leisure time despite working longer hours, attributing this to efficiency in household production. Gimenez-Nadal and Sevilla (2012) find similar patterns in Spain and the UK.

However, most studies focus on total leisure time, not health-promoting activities specifically. Cutler and Lleras-Muney (2010) show education predicts health behaviors like exercise, smoking cessation, and preventive care—suggesting education gradients in well-being time may have long-term health consequences.

2.2 Gender, Education, and Time Use

Research on gendered returns to education focuses primarily on labor market outcomes. Goldin (2006) documents the gender pay gap narrows with education, while Blau and Kahn (2017) show returns to education differ by gender and family status. However, few studies examine education’s effects on non-market time use.

For India specifically, Chatterjee et al. (2018) show educated women have higher labor force participation but also face greater household work burdens when employed—suggesting education may increase total work rather than leisure. Desai and Andrist (2010) document that educated Indian women face similar mobility restrictions and domestic responsibilities as less-educated women when married.

2.3 Health Behaviors and Social Stratification

Link and Phelan (1995) propose that socioeconomic status operates as a “fundamental cause” of health inequality because high-SES individuals use resources to avoid health risks across changing contexts. Education may facilitate health-promoting time use through: (1) knowledge of health benefits, (2) financial resources to access facilities, (3) neighborhoods with safe exercise spaces, and (4) social norms privileging fitness.

Pampel et al. (2010) show education gradients in health behaviors have widened over time in developed countries. For developing countries, Cutler et al. (2015) document emerging

education-health gradients in India, China, and Africa as health knowledge diffuses first to educated populations.

2.4 Marriage, Location, and Time Constraints

Marriage affects time allocation differently by gender. Married women perform more household work than single women while married men perform less than single men (Bianchi et al. 2000). If marriage imposes time constraints that overwhelm education’s benefits, we would observe conditional education effects.

Geographic location matters for leisure infrastructure. Urban areas provide parks, gyms, cultural venues that facilitate personal well-being activities. If education’s returns depend on accessing this infrastructure, urban-rural divides will condition education effects.

3 Data and Methods

3.1 Data Source

India’s Time Use Survey 2024 (described in Data section above) provides comprehensive 24-hour time diaries. We use a 10% random sample (N 840,000 person-days after restrictions) with adjusted survey weights.

3.2 Measuring Personal Well-Being Time

We define personal well-being time as time spent on activities enhancing personal development, health, social connections, and recreation (ICATUS codes detailed above). We separately analyze:

- **Exercise time** (codes 831-832): Physical exercise, sports
- **Social time** (codes 711-719): Visiting friends, socializing
- **Cultural time** (codes 821-829): Arts, hobbies, entertainment

3.3 Education Measurement

We use three education measures:

1. **Three-category classification:** Primary or less, Secondary, Higher Secondary+ (for descriptive analysis)
2. **Binary higher education indicator:** Higher Secondary+ vs. below (for regressions)
3. **Years of education:** Imputed from education categories (for gradient analysis)

3.4 Analytical Approach

We estimate education effects using weighted linear regression: